

1. How we can connect Armstrong's Inference rules to derive new functional dependencies from the given one
2. Draw an EER diagram for the firm where Staff is the superclass that contains (S-ID, Name, Salary) and Staff's are specialized into Project Manager and Developers ensuring each staff should be in at least one of the specialized subclass
3. Differentiate B and B+ tree
4. Differentiate DTD and XML schema
5. A relation $R(A,B,C)$ can be decomposed into $R_1(A,B)$, $R_2(B,C)$ and $R_3(A,C)$. How can you tell whether this represents join dependency?
6. Explain Lock Granularity
7. Distributed transactions happening in site X and site Y, where transaction T updates both sites. Explain the various steps of Two Phase Protocol and analyze what happens if one site fails during the commit phase
8. Explain the levels of Data Abstraction
9. Explain the importance of DBA in establishing the smooth working of database
10. Explain and suggest a solution for the database system face bucket overflow in hashing.
11. Explain and identify the CAP properties involved in distributed database system

12. The banking system deals displays incorrect account balances when two users update the same record. Explain about the issue and suggest a concurrency control mechanism that deals with it

13. Analyse the need of Normalization? What problem does it solve?

14. Develop an ER to relational mapping for a system with weak entity dependent related to student, including all constraints

15. A database contains relations, Client(CID, Name), Orders(OID,CID,Amount). Analyse how relational algebra operations can be combined to find the clients who have placed orders above 5000

16. Compare static hashing and dynamic hashing?

17. Consider the relation

Supply_part(Supply_ID,Supply_Name,Part_ID,Part_Name,Amount) Functional Dependencies are Supply_ID→Supply_Name,Part_ID→Part_Name, (Supply_ID,Part_ID)→Amount . Identify the highest Normal form

18. Compare homogeneous and heterogeneous databases with examples?

19. What is a deadlock? Explain how it can be handled?

20. Transactions:

- a. T1 locks A→ requests B
- b. T2 locks B→ requests A
 - i. How deadlock happens
 - ii. Draw wait for graph

iii. Explain deadlock prevention mechanism

21. Explain in detail about the basic inference rules that are used to derive the functional dependency

22. Explain the Wait-Die and Wound-Wait timestamp based schemes.

23. What are the different file organization methods? Explain any two?

24. Explain Strict Two Phase locking and analyze how its better than the basic one

25. Explain how RAID improves the performance and data loss

26. Describe CAP Theorem and explain how it applies to distributed databases?

27. What is distributed transactions? Explain briefly?

28. Explain database recovery techniques and the role of the transaction log?

29. How B tree and B+ tree play a major role in indexing. Explain in detail

30. List out how failure occurs during the failure of primary nodes with the help of replica set architecture

31. Given a relation **R(A, B, C, D)** with functional dependencies **A → BC** and **B → C**, find all the candidate keys of the relation.

32. Design a DTD for the XML related to a book explain in detail about each items

33. Describe the architecture of a database system with a neat diagram?

34. Describe about the ER to Relational Mapping with good examples

35. Explain the Entity-Relationship (E-R) model and its components?

36. Describe the process of normalization up to Third Normal Form (3NF) with suitable examples?

37. A file contains 5000 records and has no index.

- a. How many blocks needed to perform linear search
- b. If an index is created on the file compare the number of blocks accesses required for searching with and without the index

38. List out the various levels of RAID and explain in detail

39. Consider a relation **R(A, B, C, D)** with functional dependencies **A → BC** and **B → C**.

- a. Find the candidate key(s).
- b. Determine whether the relation is in **3NF** and **BCNF**.
- c. If not, decompose it into relations that satisfy BCNF.

40. Describe the object identity and reference types in SQL with example and also make a comparison with relational database

41. What is functional dependencies, inference rules and minimal set of functional dependencies?

42. Describe the role of Concurrency Control with Locking Methods?

43. The schedule :

- a. T1: Lock-X(A),Read(A),Write(A),Unlock(A)
- b. T2: Lock-X(A),Read(A),Write(A),Unlock(A)\
 - i. The schedule follows 2PL or Not
 - ii. Check if it is serializable

44. Consider the relation

Employee(Emp_ID,Emp_Name,Dept_ID,Dept_Name,Trainer_ID,Trainer_Phone)

Functional Dependencies are (Emp_ID→Emp_Name, Dept_ID→Dept_Name, Trainer, Trainer→Trainer_Phone).Normalize the relation upto 3NF